

Pratham Arora

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Education

Plaksha University

B.Tech. in Computer Science & AI

Mohali, India | Expected Aug. 2026

CGPA: 8.20 / 10.0

Skills

Languages: Python, SQL, JavaScript, TypeScript

Web & Backend: FastAPI, Node.js, Next.js (SSR/CSR), REST APIs, Tailwind CSS

Cloud & Tools: Azure (Cosmos DB, OpenAI), Firebase, FAISS, DiskANN, Git/GitHub, Vercel

AI/ML: PyTorch, LangChain, RAG Pipelines

Industry Experience

AI Engineering Intern

June 2025 – July 2025

Cotality (PropTech SaaS) | Kolkata

- Architected **FastAPI** backend parsing 50,000+ lines of code daily across 15+ programming languages via Abstract Syntax Tree (AST) analysis, enabling automated real-time documentation generation across polyglot codebases.
- Designed and delivered 12 production **RESTful API endpoints** with typed Pydantic response schemas, powering real-time code documentation preview consumed by the client-facing interface from day one of deployment.
- Reduced AI embedding generation costs by **85%** by designing a hash-based change detection system with **SHA-256 on AST nodes**, skipping unchanged functions to eliminate redundant vector database writes.
- Built **RAG pipeline** using LangChain, Azure OpenAI embeddings, and Cosmos DB with DiskANN vector search; migrated from FAISS as retrieval latency requirements tightened with codebase growth.

Software Development Engineer Intern

June 2024 – July 2024

Orangewood Labs (Robotics) | Noida

- Designed modular LLM task-planning architecture for a food-preparation robotic arm, separating high-level intent logic from low-level robot actuation and establishing clean integration boundaries for parallel team workflows.

Technical Projects

VR LLM Conversational Agent | Unity, Gemini API, Google Cloud STT/TTS

Aug. 2025 – Dec. 2025

- Engineered a real-time VR conversational agent integrating **Gemini 2.5 Flash** with Google Cloud STT/TTS, achieving average response latency of 1.8s through parallel API calls, caching, and audio preprocessing.
- Designed and ran a **within-subjects user study** ($N = 18$) comparing three latency feedback mechanisms; **co-authored paper submitted to MIT's Presence journal** (under review).

Anuj Desai Associates – Company Website | Next.js, Firebase, Google OAuth

2025

- Designed and shipped a full-stack site for a CA firm with a careers page and a private admin dashboard where firm partners review and manage candidate applications, powered by **Firebase Realtime Database**.
- Implemented Google OAuth via **Firebase Authentication** with email-allowlist access control, restricting dashboard access to authorised partners while keeping the public application form accessible.
- Structured the codebase for non-technical maintainability, with content editable directly via GitHub without touching component logic, eliminating the client's ongoing developer dependency.

AI Resume Builder & Interview Prep Tool | Next.js, Tailwind, Firebase, Gemini API

Jan. 2025 – Apr. 2025

- Built a full-stack AI tool with real-time PDF preview and ATS-optimised resume generation; integrated **Gemini 2.5 Flash** to generate personalised interview questions by cross-referencing user resumes with a DSA/STAR prep database, with **Firebase Authentication** and Google OAuth for user management.

Leadership & Involvement

Head of Technology

Apr. 2025 – Feb. 2026

Athleda – Plaksha Sports Society | Plaksha University

- Built and maintained a sports platform serving **500+ students**, shipping event registration, live-scoring, and photo galleries, while mentoring 2 junior developers through code reviews and production deployments.

LEAP.AI @ Plaksha

Feb. 2024 – Feb. 2025

Web Developer → Head of Web Development | Plaksha University

- Led a team of 7–8 through 2 full redesigns as Head, growing the AI club platform to **800+ monthly users**.